

Internship Final Report

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Major: Computer Science
Internship Duration: April 1st, 2026 - April 30th, 2026
Company: ShadowFox
Domain: Data Science
Mentor: Mr. Aakash
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Objectives

The main goal of this internship was to learn the basics of data science and get some real-world experience working with real datasets. The internship was all about helping me get better at data preprocessing, exploratory data analysis, visualization, and drawing useful conclusions.

The goals were:

1. Understanding the whole process of data analysis, from raw data to insights.
2. Getting real-world experience with Python, Pandas, Matplotlib, and Seaborn.
3. Using structured datasets to clean and prepare data for analysis.
4. Looking at environmental and business datasets to find patterns and trends.
5. Making it possible to show results using reports and charts.

Tasks and Responsibilities

During the internship, I worked on three levels of tasks: Beginner, Intermediate, and Advanced.

Beginner Level Task:

At the beginner level, I focused on learning the basics of how to handle and preprocess data.

- Used Pandas to import datasets and look at how they were set up
- Dealt with missing values and cleaned up data that didn't match.
- Changed the types of data and set up the columns correctly.

- Did basic statistical analysis
- Created simple graphs and charts, like bar charts and line plots.

This step helped me get a good start on working with datasets and figuring out how they are put together.

Intermediate Level Task: Analysing the AQI in Delhi

For this task, I looked closely at the Air Quality Index (AQI) data for Delhi.

1. Set research questions about pollution levels and how they change with the seasons.
2. Cleaned up and prepared environmental data.
3. Figured out AQI values based on how much pollution there was.
4. Looked at pollutants like PM_{2.5}, PM₁₀, and NO₂.
5. Looked at seasonal patterns (Winter, Summer, Monsoon, and Post-Monsoon).
6. Made pictures like these:
7. Line plots show how the AQI has changed over time.
8. Box plots show how things change with the seasons.
9. Bar charts (AQI levels by month).
10. Heatmaps (correlations between pollutants).

This task helped me get better at looking at environmental data and figuring out problems like pollution in the real world.

Advanced Level Task: Analyzing Amazon Sales Data

I used Jupyter Notebook to do a full analysis of an Amazon sales dataset for the advanced task.

1. Looked at the dataset and figured out how it was put together.
2. Data was cleaned and preprocessed (dealing with missing values and formatting columns).
3. Did feature engineering, like converting dates and getting the month out of a date.
4. Set research questions about sales trends and how well they do.
5. Made a lot of different visualizations, such as:
6. Sales trends over time.
7. Comparison of sales by month.
8. Sales by category.
9. Best-selling items.
10. Distribution of sales (boxplot, histogram).
11. Geographical analysis (sales by state and city).
12. A heatmap of correlations.
13. Insights gained to better understand how a business is doing and how customers act.

This task helped me get better at analyzing and visualizing business data in the real world.

Learning Outcomes

I learned a lot through this internship:

1. Skills in technology:

- Learned how to use Python libraries like Pandas, Matplotlib, and Seaborn well.
- Learned how to clean, prepare, and change data.
- Learned a lot about data science workflow and how to visualize and analyze data well.
- Learned about the whole process of data analysis.
- Learned how to turn raw data into useful information.
- Learned how to define and answer research questions.

2. Thinking in an analytical way:

- Better at finding patterns, trends, and connections in data.
- How to read visualizations well.
- Learned how to use data to solve problems.

3. Skills for Work:

- Better skills in writing reports and keeping records.
- Learned how to make findings clear.
- Learned how to manage time and get things done.

Challenges and Solutions

1. Problems with Cleaning Data:

Some datasets had missing values, formats that didn't match, and wrong entries.

Answer:

Used preprocessing methods like getting rid of null values, making sure all columns are in the same format, and fixing data types.

2. Working with Date Formats:

Different date formats made it hard to analyze.

Answer:

Used the right methods for parsing dates in Pandas and dealt with bad values using error handling techniques.

3. Mistakes in Visualization:

There were mistakes because the column names were wrong and some data was missing.

Answer:

Before plotting, I used `df.columns` to carefully check the dataset columns and make sure they were being used correctly.

4. Getting to Know Big Datasets:

It was hard to understand at first to look at more than one variable at a time.

Answer:

Divided the analysis into smaller parts and worked on one variable at a time.

Conclusion

The internship at ShadowFox gave me invaluable practical data science experience. I was able to hone my skills in data pretreatment, analysis, and visualization through organized activities.

My comprehension of real-world environmental data was enhanced by the intermediate AQI analysis, my ability to work with business statistics was reinforced by the advanced Amazon sales analysis, and the beginner tasks helped me grasp the fundamentals.

All things considered, this internship improved my technical expertise, critical thinking skills, and self-assurance when working on data science projects. It has also inspired me to study more complex data science issues and seek a job in this area.

Acknowledgments

I want to sincerely thank ShadowFox for giving me this great internship opportunity. I am appreciative of the organized exercises and educational materials that improved my data science abilities.

I particularly value the advice and assistance I received during the internship, which really enhanced my educational experience. This internship has been crucial to my professional and academic growth.